

Name of Course	: CBCS B.Sc. (H) Mathematics
Unique Paper Code	: 32351302_OC
Name of Paper	: C6-Group Theory-I
Semester	: III
Duration	: 3 hours
Maximum Marks	: 75 Marks

Attempt any four questions. All questions carry equal marks.

1. Let $G = \text{GL}(n, \mathbb{R})$. Let $H = \{A \in G \mid \det A \text{ is a power of } 5\}$. Then prove or disprove that H is a subgroup of G . Find the elements in $U(10)$ and $U(12)$ that satisfy the equation $x^2 = 1$.
2. List all the elements of order 3 in $\mathbb{Z}_{24}$. Find the smallest subgroup of $\mathbb{Z}$ containing 12 and 18. Determine the subgroup lattice for $\mathbb{Z}_{24}$.
3. Let S_n be the symmetric group of degree n . Suppose that $\alpha \in S_n$ can be written as a product of disjoint cyclic permutations of lengths $m_1, m_2, \dots, m_r$, ($r \in \mathbb{N}$), respectively. Then prove that the order of α is $\text{lcm}(m_1, m_2, \dots, m_r)$. Find the orders of $(13)(27)(456)(8)(1237)(648)(5)$ and $(124)(345)$. Furthermore, show that if H is a subgroup of S_n then either every member of H is an even permutation or exactly half of them are even. Also, find $Z(S_n)$ for $n \geq 3$.
4. Show that for a finite group G , the index of a subgroup H in G is $|G|/|H|$. Prove that every subgroup of index 2 of a group G is normal. Give an example of a subgroup H of index 3 in a group G which is not normal in G . Also, determine the index of $3\mathbb{Z}$ in $\mathbb{Z}$.
5. Let $H = \{\beta \in S_5 : \beta(1) = 1\}$ and $K = \{\beta \in S_5 : \beta(2) = 2\}$. Prove that H is isomorphic to K . Is the same true if S_5 is replaced by S_n , where $n \geq 3$? Further prove or disprove that S_4 is isomorphic to D_{12} .
6. If H is a subgroup of G and K is a normal subgroup of G , then prove that $H/(H \cap K)$ is isomorphic to HK/K . Also determine all homomorphisms from $\mathbb{Z}_n$ to itself.

Name of Course : CBCS B.Sc. Hons Mathematics
 Unique Paper Code : 32351302
 Name of Paper : BMATH306-Group Theory-1
 Semester : III
 Duration : 3 hours
 Maximum Marks : 75 marks

Attempt any four questions. All questions carry equal marks.

1. Let A be a non-empty set and $\langle G, . \rangle$ be a group. Let F be the set of all functions from A to G . Define an operation $*$ on F as follows:

For $f, g \in F$, let $f * g : A \rightarrow G$ as $(f * g)(x) = f(x) \cdot g(x) \forall x \in A$.

Prove that $\langle F, * \rangle$ is a group.

Find the inverse of $\begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$ in $GL(2, \mathbb{Z}_5)$, the group of 2×2 non-singular matrices over $\mathbb{Z}_5$. Verify the answer by direct calculation.

Describe the group of symmetries of a non-square rectangle and draw its Cayley's table.

2. Let a be an element of a group such that $|a| = 3$, prove that $C(a) = C(a^2)$. Give an example to show that the conclusion fails if $|a| = 4$.

Find the orders of each of the elements of $U(14)$. Show that it is cyclic and find all its generators.

3. Define Centre $Z(G)$ of a group G and prove that $Z(S_4) = \{e\}$.

For $n > 2$, show that every even permutation in S_n is a product of 3-cycles.

Let $\sigma = (1,5,7)(2,5,3)(1,6)$. Express σ^{17} as a cycle.

4. Prove or disprove any six, stating the results used

(i) $\langle \mathbb{R}, + \rangle \approx \langle \mathbb{Q}, + \rangle$, (ii) $\langle \mathbb{Q}, + \rangle \approx \langle \mathbb{Z}, + \rangle$, (iii) $\langle \mathbb{R}, + \rangle \approx \langle \mathbb{R}, . \rangle$,

(iv) $D_4 \approx$ Group Q of Quaternions, (v) $U(20) \approx D_4$,

(vi) $U(8) \approx U(12)$, (vii) $U(10) \approx \mathbb{Z}_4$, (viii) $\frac{GL(2, \mathbb{R})}{SL(2, \mathbb{R})} \approx \mathbb{R}^*$.

5. Let H be a subgroup of a group G . Prove that $aH \mapsto Ha^{-1}$ is a bijective mapping from the set of all left cosets of H in G to the set of all right cosets of H in G . Can the same be said for $aH \mapsto Ha$?

If G is a non-abelian group of order 8 with $Z(G) \neq \{e\}$, prove that $|Z(G)| = 2$.

6. Let N be a normal subgroup of G and M be a normal subgroup of N . If N is cyclic, prove that M is a normal subgroup of G . Show by an example that the conclusion fails to hold if N is not cyclic.

If φ is a homomorphism from a finite group G to a finite group G' , prove that $|\varphi(G)|$ divides the gcd of $|G|$ and $|G'|$.